Test Report

nanosecond triple-wavelength pulsed laser

Product Name:	**nanosecond triple-wavelength pulsed laser**		
Model:	**Grandark-NTPL-266/532/1064**	**Serial No.:**	**202504001**
Inspection Engineer No.:	**2020120**	**Inspection Date**	**April 10th, 2025**

1. Test Environment

Ambient Temperature: 23.2 ℃, humidity: 46 %

2. Power Test of Laser Output

2.1 Test at 1MHz

Test conditions: Working frequency at 1MHz, pulse width at 2ns

Current (mA)	1064nm Output Power (W)	532nm Output Power (W)	266nm Output Power (W)
0	2.69	0.34	/
1000	4.86	0.92	/
1500	8.32	2.23	/
2000	12.15	4.13	0.07
2500	16.01	6.51	0.17
3000	19.78	8.86	0.38
3500	/	/	0.72
4000	/	/	1.11
4500	/	/	1.47
5000	/	/	2.29

2.2 Test at different frequency

Test conditions: pulse width at 2ns

2.2.1 1064nm Output Power

Repetition Frequency (kHz)	100	200	300	400	500	600	800	1000	1200	1400	1600	1800	2000
Current (mA)	1300	1800	2200	2800	3000	3000	3000	3000	3000	3000	3000	3000	3000
Power (W)	4.47	8.39	11.32	16.27	18.79	19.59	20.38	20.95	21.05	21.14	21.16	21.24	21.33

2.2.2 532nm Output Power

Repetition Frequency (kHz)	100	200	300	400	500	600	800	1000	1200	1400	1600	1800	2000
Current (mA)	1300	1800	2200	2800	3000	3000	3000	3000	3000	3000	3000	3000	3000
Power (W)	1.02	3.79	5.23	9.15	10.85	10.40	10.54	10.13	9.69	9.35	8.86	8.53	8.10

2.2.3 266nm Output Power

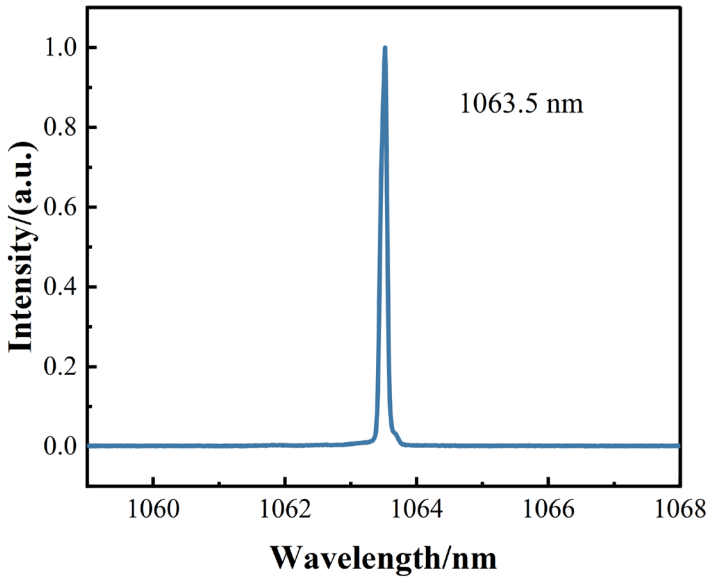
Repetition Frequency (kHz)	100	200	300	400	500	600	800	1000	1200	1400	1600	1800	2000
Current (mA)	1300	1800	2200	2800	3100	3500	4400	5000	5000	5000	5000	5000	5000
Power (W)	0.02	0.15	0.25	0.48	0.75	1.01	1.49	2.29	1.71	1.51	1.33	1.11	0.94

3. Spectrum Test

Test conditions: current 5.5A, working frequency 1MHz, pulse width at 2ns

3.1 1064nm Spectrum Test

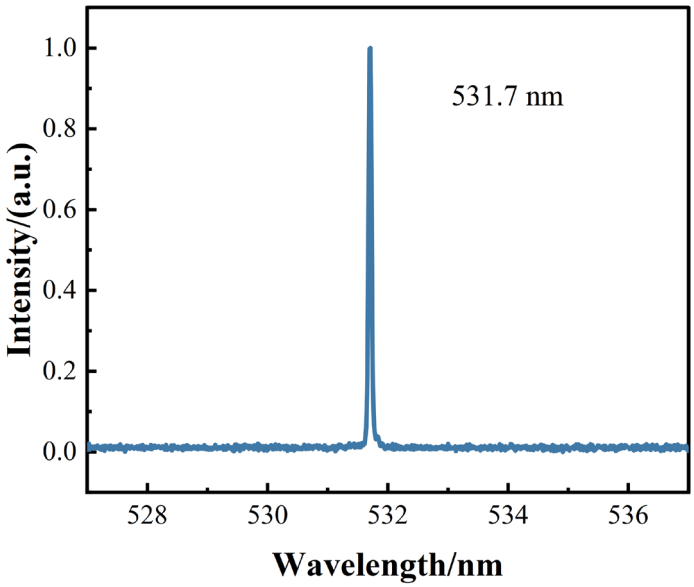
Test result: center wavelength is 1063.5 nm



Picture 1. 1064nm Spectrum Test

3.2 532nm Spectrum Test

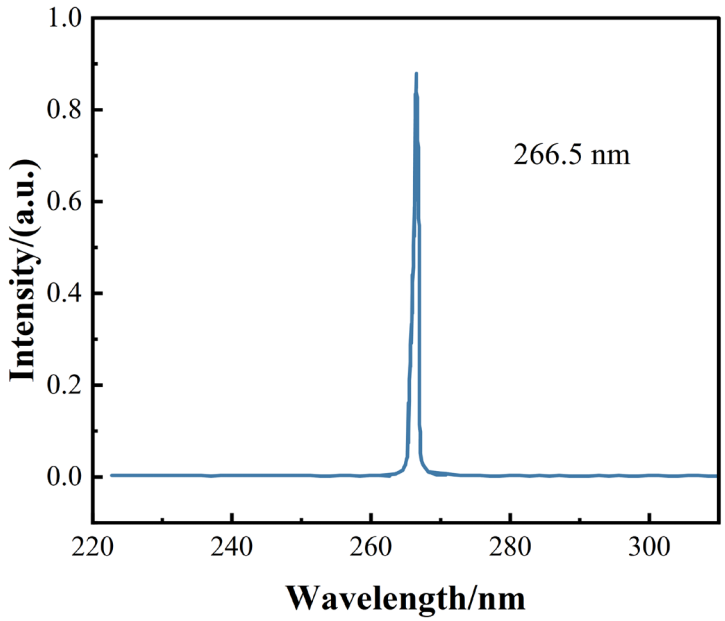
Test result: center wavelength is 531.7 nm



Picture 2. 532nm Spectrum Test

3.3 266nm Spectrum Test

Test result: center wavelength is 266.5 nm



Picture 3. 266nm Spectrum Test

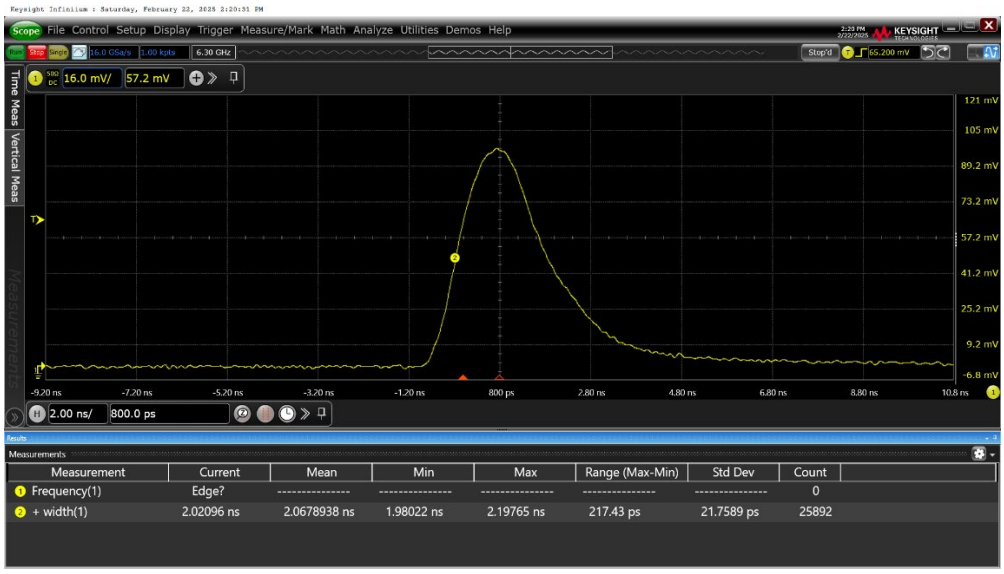
4. Test of Typical Pulse Waveform and Pulse Width

Test conditions: current 5.5A, working frequency 1MHz, pulse width at 2ns

4.1 1064nm Pulse Test

Test result: repetition frequency is 1.00 MHz, pulse width is 2.02 ns

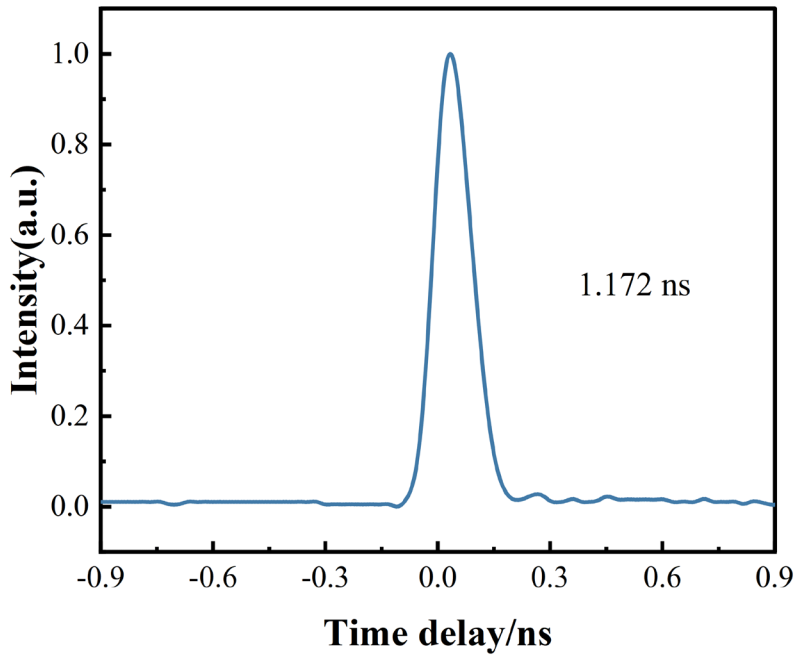




Picture 4. 1064nm Typical Pulse Waveform and Pulse Width.

4.2 532nm Pulse Test

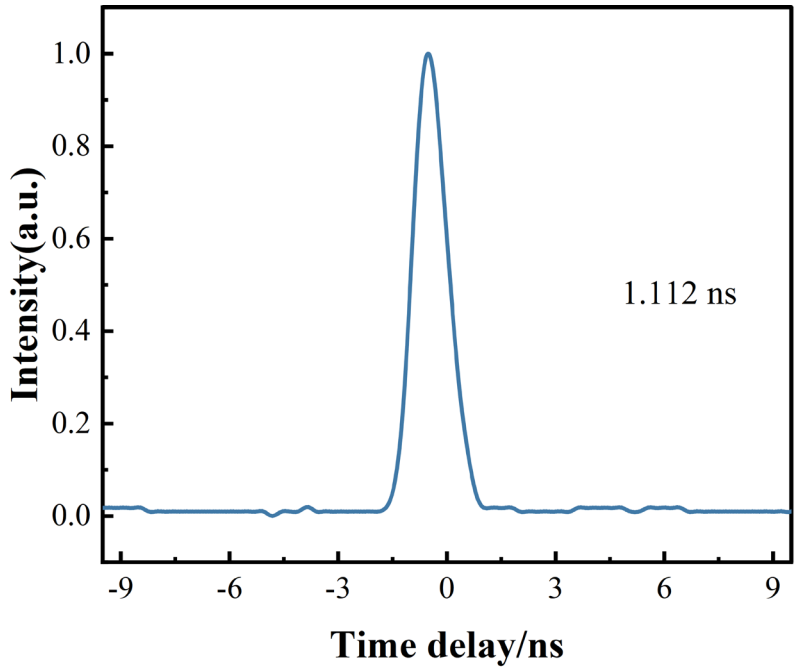
Test result: pulse width is 1.172 ns



Picture 5. 532nm Typical Pulse Waveform and Pulse Width.

4.3 266nm Pulse Test

Test result: pulse width is 1.112 ns

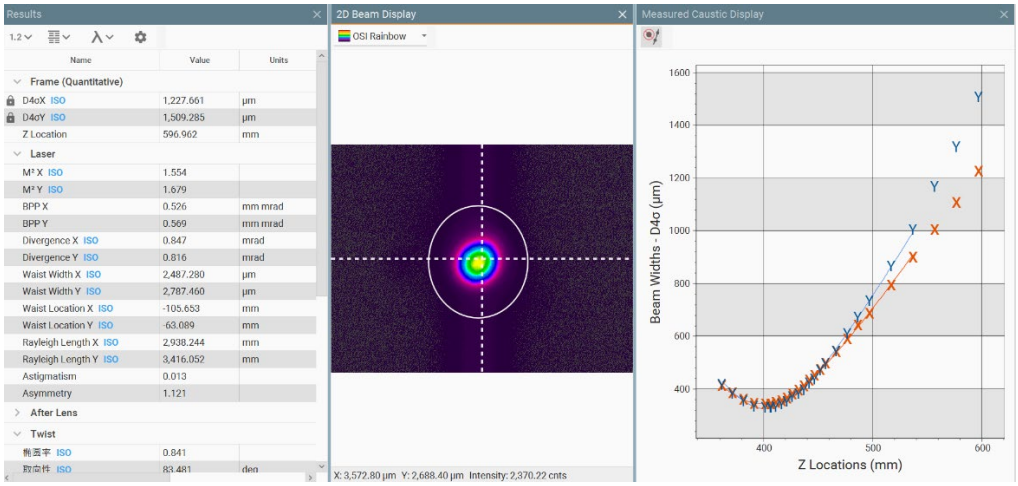


Picture 6. 266nm Typical Pulse Waveform and Pulse Width.

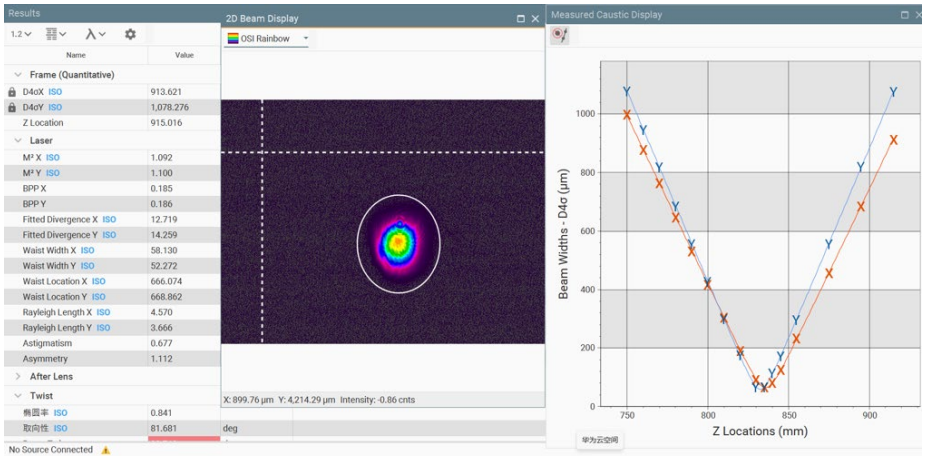
5. Beam Test

5.1 Beam Quality Test

Test result: current 5.5A, working frequency 1MHz, pulse width is 2ns



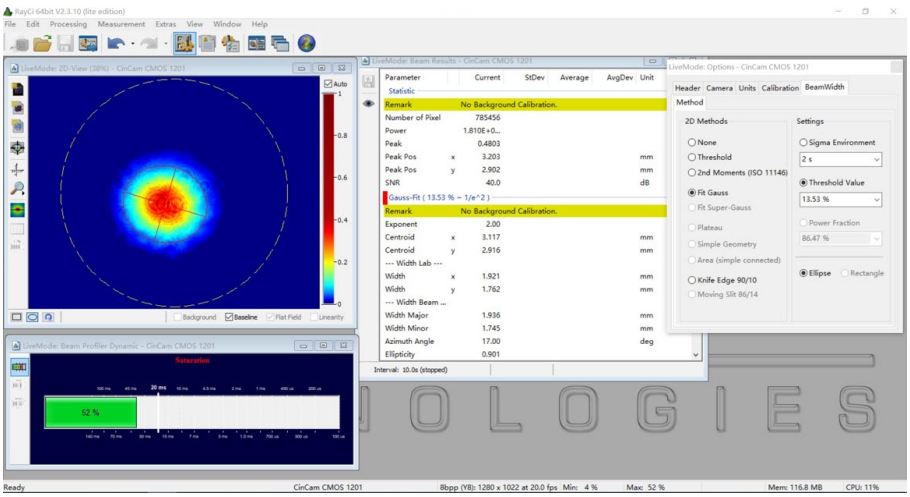
Picture 7. 1064nm beam quality test (M^2X=1.554 M^2Y=1.679)



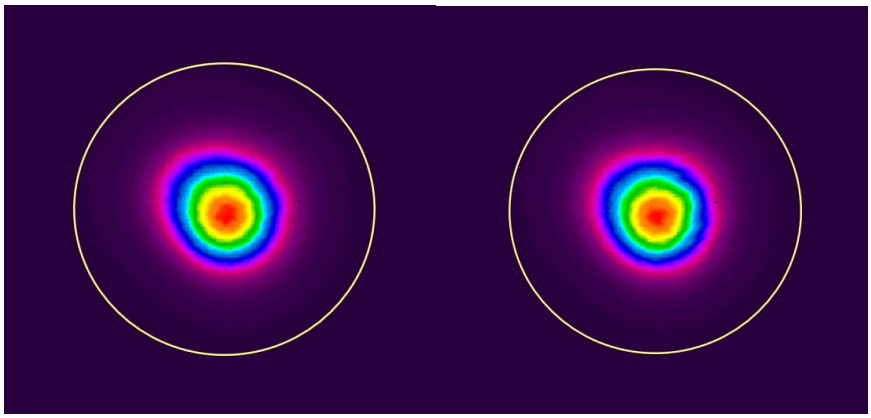
Picture 8. 532nm beam quality test (M^2X=1.092 M^2Y=1.100)

5.2 Near-field Spot test

Test conditions: current 5.5A, working frequency at 1MHz, pulse width at 2ns



Picture 9. 1064nm near-field spot test



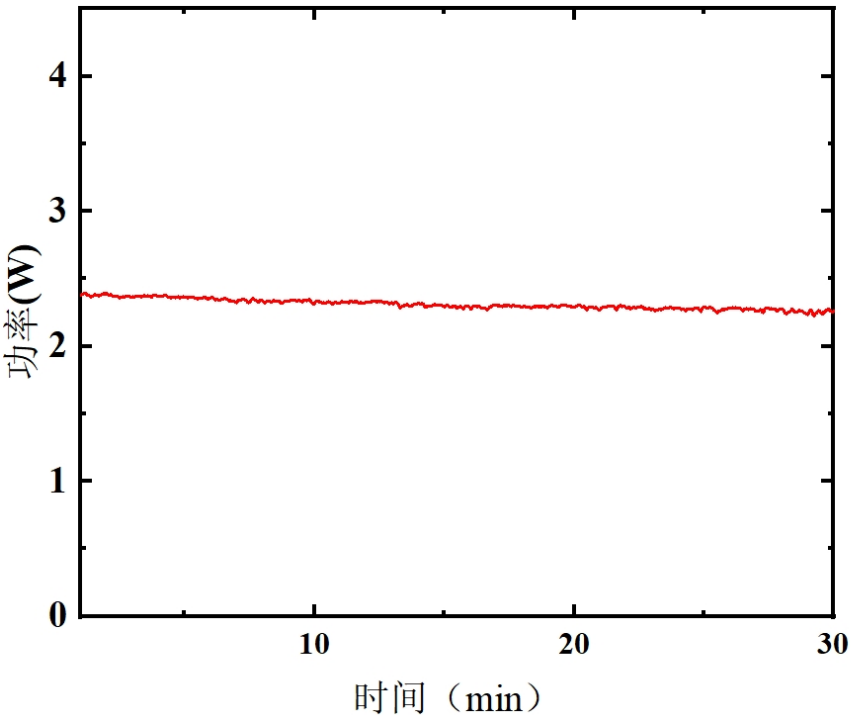
Picture 10. 532nm near-field spot test

Picture 11. 266nm near-field spot test

6. Power Stability Test

Test conditions: current 5.5A, working frequency at 1MHz, pulse width at 2ns

Test results: average power 2.313W, RMS stability 1.633%



Picture 12. 266nm power stability test